Dashboard for EcoPlus Productivity

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Background

Ecoplus International specialising in piping and engineering for the construction, marine, and oil & gas industries. They provide 3 core piping services — Threading, Grooving and Welding.

Problem

- Dashboard lacks clarity and intuitive navigation difficult to interpret key information
- Limited visibility into project and workforce metrics impacts efficiency and decision-making

Objective

Enhance workforce visibility and improve manpower planning and operational decision-making.

- Consolidate order and work data into structured sheets
- Develop a web app for streamlined data entry
- Build Power BI data visualisations for real-time work tracking

Methodology & Tools

Data Exploration
- Data cleaning & preprocessing
- Consolidated scattered data into one format

Data Filter
- Identified and Filtered important data

Data Visualisation
- Designed intuitive data visualizations to present relevant insights clearly

Dashboard Ideation
- Brainstormed data entry methods and dashboard variations

Refinement
- Iterated on dashboard and form designs to enhance user experience

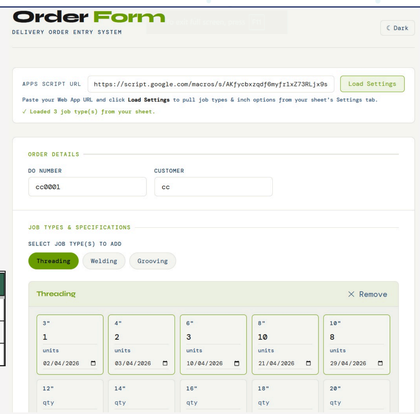
Main Result

1 Order Form

Enables staff to submit delivery orders with job types and specifications, automatically recorded into the dataset

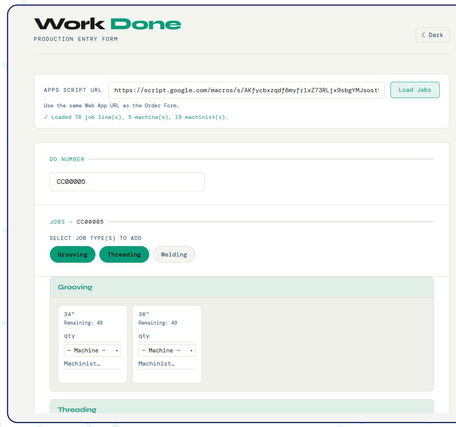
DO Number	Job Type	Inches	# Quantity	Date
DO-0001	Welding	20"	1	2026-03-26
CC-123456	Threading	16"	8	2026-03-27
CC-123456	Threading	16"	1	2026-03-27

Shift	Machinist	Machine	Submitted At	Estimated Manhours Taken
Day	Test1	Manual 1	3/26/2026, 2:27:00 PM	2.8
Day	mary	CNC 3	27/03/2026, 19:02:09	14.4
Day	joanne	CNC 3	27/03/2026, 19:05:19	1.8



2 Work Done

Allows workers to log completed work against existing orders, automatically updating remaining quantities upon submission



DO Number	Customer	Required Date	Job Type
CC000001	NewOrderT...	2026-03-28	Threading
CC000001	NewOrderT...	2026-03-28	Threading
CC000001	NewOrderT...	2026-03-28	Threading

Inches	# Quantity	Submitted At	Covert Text to Date
3"	1	28/03/2026	2026-03-28
4"	1	28/03/2026	2026-03-28
6"	1	28/03/2026	2026-03-28

3 Job Status

DO Number	Customer	Required Data	Job Type	Inches	Quantity	Date Submitted	Estimated Manhours Needed	Quantity Completed	Quantity Remaining	Estimated Manhours Left	Status
CC000001	NewOrderTest1	2026-03-28	Welding	12"	1	2026-03-28	1	1	0	0	Completed
CC000010	TestDO	2026-03-31	Threading	3"	1	2026-03-31	0.4	0	1	0.4	In Progress
CC000009	Test9	2026-05-20	Grooving	26"	10	2026-03-28	43	0	10	43.00	In Progress
CC00005	NewOrderTest5	2026-12-31	Welding	34"	50	2026-03-28	315	1	49	308.7	In Progress

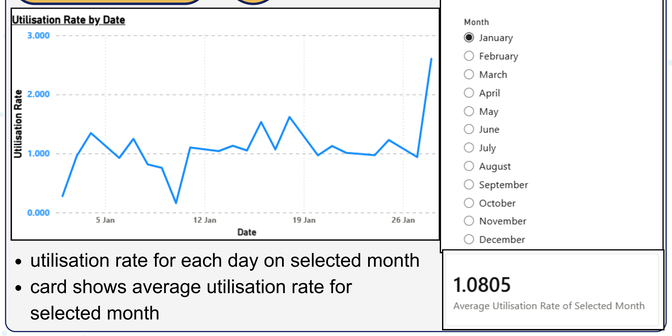
- Provides an at-a-glance view of all orders with completed and remaining quantities, colour-coded by urgency:
  - Green — Completed
  - Red — Overdue
  - Yellow — Due within a month
  - White — Not due soon

4 Dashboard

- Compares on-time vs overdue quantities by job type, enabling quick identification of delays
- Capacity and utilisation metrics reflect how effectively the workforce was deployed each day

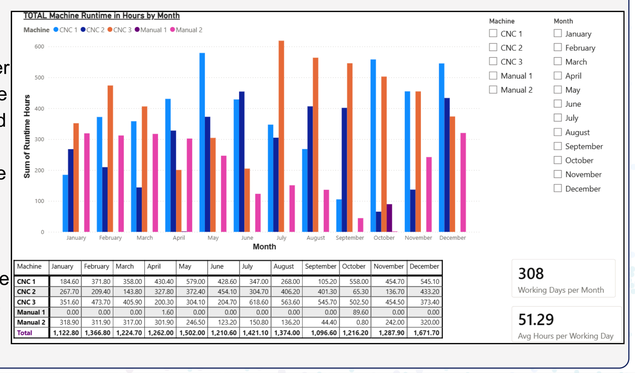
Date	O. Thread (1)	O. Thread (2)	O. Grve (1)	O. Grve (2)	O. Weld (1)	O. Weld (2)	O. Total	C. Threading (M	C. Grooving (MH	C. Welding (MH	C. Total	Capacity (MH)	Utilization
2026-03-28	92	661.8	46.4	769	22.1	698	2289.3	25.8	12.6	30.3	68.7	30	2.29
2026-03-29	78.9	658.3	37.9	769	12.6	692.5	2249.2	4.8	2.1	14.7	21.6	140	0.1542857143
2026-03-30	120	652.5	74.8	765.1	52.4	698	2362.8	11.7	11.3	12.6	35.6	50	0.712

5 Power BI Visualisation

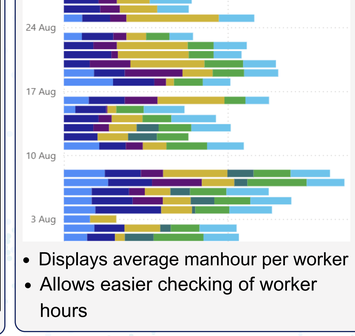


- utilisation rate for each day on selected month
- card shows average utilisation rate for selected month

- Displays total runtime hours per machine per month, filterable by machine and month
- Cards and table provide a breakdown of working days, average runtime and total tallies per machine



Machine	January	February	March	April	May	June	July	August	September	October	November	December
CNC 1	184.60	371.80	358.00	430.40	579.00	428.60	347.00	268.00	105.20	558.00	454.70	545.10
CNC 2	267.70	209.40	143.80	327.80	372.40	454.10	304.70	406.20	401.30	65.30	136.70	433.20
CNC 3	351.60	473.70	405.90	200.30	304.10	204.70	618.60	563.60	545.70	502.50	454.50	373.40
Manual 1	0.00	0.00	0.00	1.60	0.00	0.00	0.00	0.00	0.00	89.60	0.00	0.00
Manual 2	318.90	311.90	317.00	301.90	246.50	123.20	150.80	136.20	44.40	0.80	242.00	320.00
Total	1,122.80	1,366.80	1,224.70	1,262.00	1,502.00	1,210.60	1,421.10	1,374.00	1,096.60	1,216.20	1,287.90	1,671.70



Date	Sum of Worker A
2025-08	0.35
2025-08	0.35
2025-08	0.53
2025-08	0.55
2025-08	0.56
2025-08	0.71
2025-08	0.76
2025-08	0.80
2025-08	0.85
2025-08	0.85
2025-08	0.92
2025-08	0.93
2025-08	0.96
2025-08	1.08
2025-08	1.25
2025-08	1.35
2025-08	1.40
2025-08	1.50
Total	15.70

- Displays average manhour per worker
- Allows easier checking of worker hours

Assumptions

- All jobs have identical difficulty regardless of type
- No waiting time between stages (threading, grooving, welding)

Limitations

- Dashboard does not account for machine downtime, which may skew utilization and runtime figures
- Built on limited real-world data so long-term reliability is yet to be tested

Conclusion

- By developing an integrated web app and Power BI dashboard, stakeholders can now efficiently track job progress, workforce utilisation, machine runtime and order status in real time, enabling timely and informed decision-making